

Writing better READMEs

Data bites series
Research Data Management Services
Phoebe Ayers, EECS Librarian
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<https://libraries.mit.edu/data-management/services/workshops/>

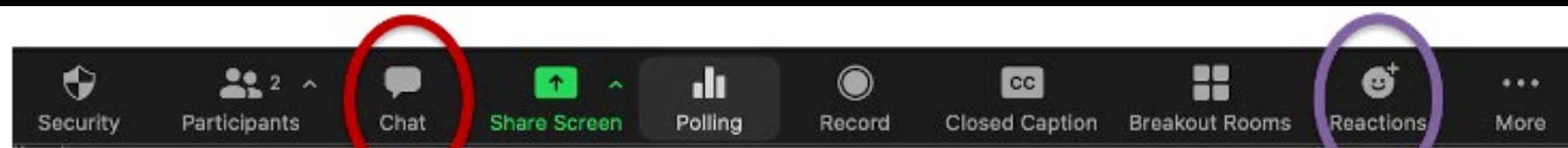
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MIT
Libraries

Ask questions

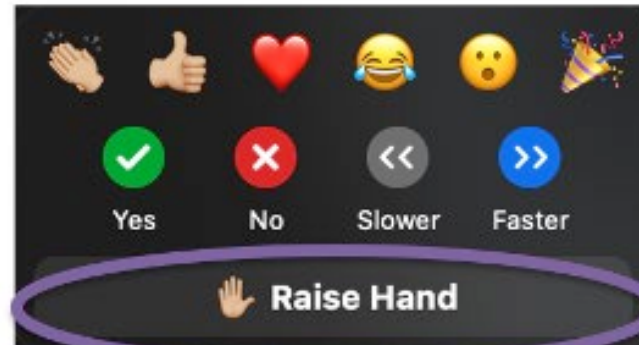
Ask questions by either

- a) typing them in the **Zoom chat window**, or
- b) use the **Raise Hand** option under the **Reactions icon** at the bottom of your Zoom window.



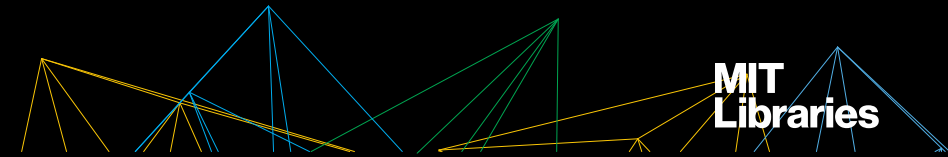
a) Zoom chat

b) Access the Raise Hand option by clicking the Reactions icon



Who is the README for?

- You!
- Your colleagues and collaborators
- Your paper reviewers
- Anyone running or using your code and data
- Future contributors to your project!

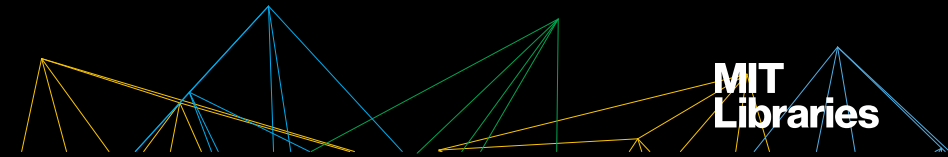


Discussion question

What are the components of your research?

Do you create...

- Analysis scripts, programs or other code?
- Datasets?
- Data analyses?
- Published articles?
- Other?



Where should you put a README?

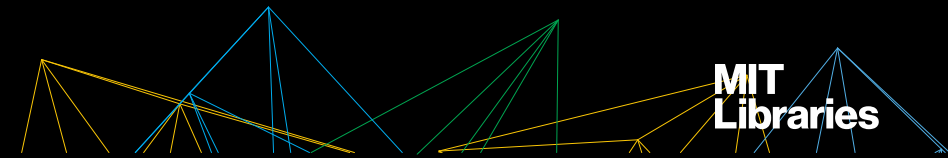
- **Data sets**

- Project-level
- Each logical “cluster” or collection of data
- Maybe: a single data file

- **Code**

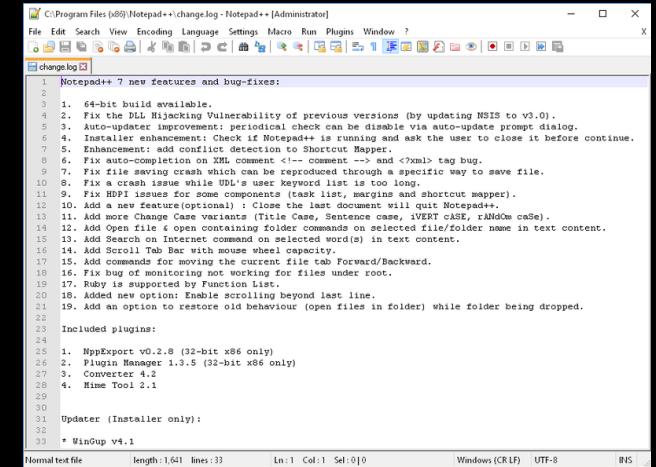
- Project-level (might link to external documentation)
- Stand-alone components (eg folders within repos on GitHub)

Make sure the README is easily associated with the described data



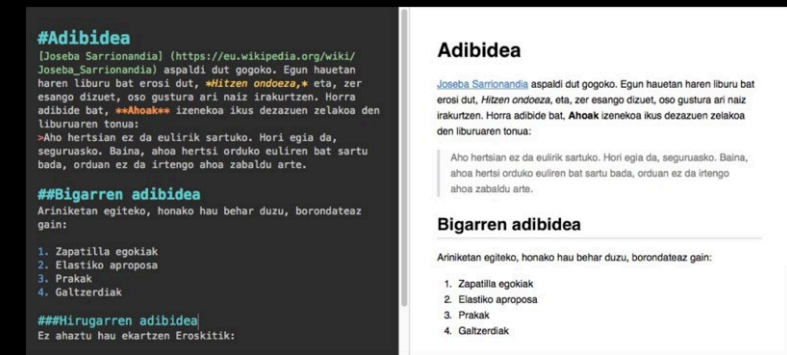
How?

- Make a text document
 - Can use any text editor: Notepad, Notepad++, atom, vim, MarkText, etc.
 - File type could be .txt or .md (markdown)
- You want it to be easily read and interoperable across systems
- GitHub uses .md by default
- Find (or make) a template for easy README creation



A screenshot of the Notepad++ application window. The title bar reads 'C:\Program Files (x86)\Notepad++\change.log - Notepad++ [Administrator]'. The menu bar includes File, Edit, Search, View, Encoding, Language, Settings, Macro, Run, Plugins, Window, and Help. The toolbar contains various icons for file operations and editing. The text area displays a changelog for Notepad++ version 7, listing new features and bug fixes. The status bar at the bottom indicates 'Normal text file', 'length: 1,641', 'lines: 33', 'Ln: 1', 'Col: 1', 'Sel: 0|0', 'Windows (CR LF)', 'UTF-8', and 'BIS'.

```
1 Notepad++ 7 new features and bug-fixes:
2
3 1. 64-bit build available.
4 2. Fix the DLL Hijacking Vulnerability of previous versions (by updating NSIS to v3.0).
5 3. Auto-updater improvement: periodical check can be disable via auto-update prompt dialog.
6 4. Installer enhancement: Check if Notepad++ is running and ask the user to close it before continue.
7 5. Enhancement: add conflict detection to Shortcut Mapper.
8 6. Fix auto-completion on XML comment <!-- comment --> and <?xml> tag bug.
9 7. Fix file saving crash which can be reproduced through a specific way to save file.
10 8. Fix a crash issue while UDL's user keyword list is too long.
11 9. Fix DPI issues for some components (task list, margins and shortcut mapper).
12 10. Add a new feature(optional) : Close the last document will quit Notepad++.
13 11. Add more Change Case variants (Title Case, Sentence case, iVERT cASE, rAlMdoM cAsE).
14 12. Add Open file & open containing folder commands on selected file/folder name in text content.
15 13. Add Search on Internet command on selected word(s) in text content.
16 14. Add Scroll Tab Bar with mouse wheel capacity.
17 15. Add commands for moving the current file tab Forward/Backward.
18 16. Fix bug of monitoring not working for files under root.
19 17. Ruby is supported by Function List.
20 18. Added new option: Enable scrolling beyond last line.
21 19. Add an option to restore old behaviour (open files in folder) while folder being dropped.
22
23 Included plugins:
24
25 1. NppExport v0.2.0 (32-bit x86 only)
26 2. Plugin Manager 1.3.5 (32-bit x86 only)
27 3. Converter 4.2
28 4. Mime Tool 2.1
29
30
31 Updater (Installer only):
32
33 * WinUpd v4.1
```



A side-by-side comparison of a README file. The left panel shows the raw markdown source code, and the right panel shows the rendered HTML output.

#Adibidea
[Joseba Sarriónandia] (https://eu.wikipedia.org/wiki/Joseba_Sarriónandia) aspaldi dut gogoko. Egun hauetan haren liburu bat erosi dut, *Hiltzen ondoeza*, eta, zer esango dizuet, oso gustura ari naiz irakurtzen. Horra adibide bat, *Ahoak* izenekoa ikus dezazuen zelakoa den liburaren tonua:
>Aho hertsian ez da eulirik sartuko. Hori egia da, seguruasko. Baina, ahoa hertsia orduko euliren bat sartu bada, orduan ez da irtengo ahoa zabaldu arte.

##Bigarren adibidea
Ariniketan egiteko, honako hau behar duzu, borondateaz gain:
1. Zapatilla egokiak
2. Elastiko aproposa
3. Prakak
4. Galtzerdiak

##Hirugarren adibidea
Ez ahaztu hau ekartzen Eroskitik:

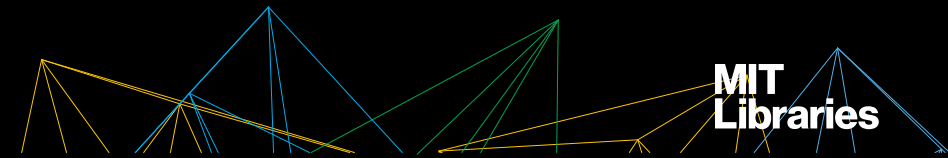
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What should you include??

All relevant info for identifying, understanding, obtaining, using and properly crediting your data and/or code

Let's break this down!



Identifying & understanding the data/code

- What is this data / code? What is it for?
- Use cases and limitations
- Related work / link to papers, if relevant

<https://physionet.org/content/deid/1.1/README.txt>

File: <base> /README.txt (10,694 bytes)

De-Identification Version 1.1 README

=====

Name:

Automated De-Identification of Free-Text Medical Records

Purpose:

This software de-identifies protected health information (PHI) from free-text medical records and outputs the de-identified text.

Identification, con't

- Explain the files and file types and their relationships
- How are they named/arranged?
- Date created, other versions

ECCO Version 4 Release 2 [\[https://dataverse.harvard.edu/dataverse/ECCOV4r2inputs\]](https://dataverse.harvard.edu/dataverse/ECCOV4r2inputs) [\[doi:10.7910/DVN/PICCRE\]](https://doi.org/10.7910/DVN/PICCRE)

- This dataset (<https://dx.doi.org/10.7910/DVN/PICCRE>) provides documentation for the 'ECCO Version 4 Release 2 Inputs' dataverse (<https://dataverse.harvard.edu/dataverse/ECCOV4r2inputs>). The contents of this dataverse allow users to re-generate ECCO V4 R2 solution, monthly output, and model-data comparisons (Forget et al. 2015, 2016) by running the MIT general circulation model (<http://mitgcm.org/>) and the included model setup (or http://mitgcm.org/viewvc/MITgcm/MITgcm_contrib/gael/verification/ECCO_v4_r2/) following the directions given in ECCOV4R2_userGuide.pdf (see <https://dx.doi.org/10.7910/DVN/ODM2IQ>).

- This dataset (<https://dx.doi.org/10.7910/DVN/PICCRE>) contains the following files:

README.pdf	general documentation of the ECCO V4 R2 inputs dataverse (this file)
c66a_eccov4r2.tar	ECCO V4 R2 model setup at the time of MITgcm checkpoint 66a

- The ECCO Version 4 Release 2 Inputs dataverse (<https://dataverse.harvard.edu/dataverse/ECCOV4r2inputs>) contains:

https://dx.doi.org/10.7910/DVN/PICCRE	documentation of the 'ECCO Version 4 Release 2 Inputs' dataverse (this file)
https://dx.doi.org/10.7910/DVN/9WYSZF	ECCO version 4 release 2 inputs: surface forcing fields
https://dx.doi.org/10.7910/DVN/7XYXSF	ECCO version 4 release 2 inputs: model initialization
https://dx.doi.org/10.7910/DVN/GNOREE	ECCO version 4 release 2 inputs: in situ T-S profiles
https://dx.doi.org/10.7910/DVN/MEDQWY	ECCO version 4 release 2 inputs: sea level anomaly
https://dx.doi.org/10.7910/DVN/L3OQT0	ECCO version 4 release 2 inputs: sea surface temperature
https://dx.doi.org/10.7910/DVN/DKXQHO	ECCO version 4 release 2 inputs: ice cover fraction
https://dx.doi.org/10.7910/DVN/F8BCRF	ECCO version 4 release 2 inputs: surface wind stress
https://dx.doi.org/10.7910/DVN/SYZMUX	ECCO version 4 release 2 inputs: bottom pressure
https://dx.doi.org/10.7910/DVN/H2Q1ND	ECCO version 4 release 2 inputs: miscellaneous

LIBRARIES

Identification: the contents

- What format are the files?
- Identify variables and null values (what does a zero mean?)
- What methods were used in creating this data?

Data Files

The `experiment-info.docx` file provides information about subjects and postures, including visual diagrams. There are also `subject-info-*.csv` files in each experiment subdirectory describing subject information.

The data files are provided in tab delimited text format, listed in subdirectories named after the experiment number and subject number.

For experiment *i*, the text files are named after the position number from 1-17. Each file contains multiple frames of recordings, with the 2048 columns representing the 32x64 pressure sensor measurements, and each row giving a new set of samples.

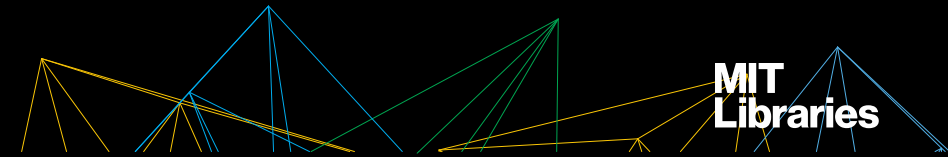
Obtaining the data / code

- Especially if restricted:
 - How can you get it?
- Other download locations, if applicable
- Contact information of the maintainers

Files

This is a restricted-access resource. To access the files, you must [sign the data use agreement](#) for the project.

<https://physionet.org/content/bigideaslab-step-hr-smartwatch/1.0/>



Using the data / code

- How to compile or install
- Software / library / package requirements
- Variables to set or APIs needed
- How can you contribute to the project??

<https://github.com/MITLibraries/timdex>

Required Environment Variables (all ENVs)

- `EMAIL_FROM` : email address to send message from, including the registration messages.
- `EMAIL_URL_HOST` - base url to use when sending emails that link back to the application. In development, often `localhost:3000`. On heroku, often `yourapp.herokuapp.com`. In production, use a custom domain in production, that should be the value you use in production.
- `JWT_SECRET_KEY` : generate with `rails secret`
- `ELASTICSEARCH_INDEX` : Elasticsearch index or alias to query
- `ELASTICSEARCH_URL` : defaults to `http://localhost:9200`

Properly crediting the work

- Who created it? Contact information?
- List any data sources / original sources
- Funding information
- Provide a citation for your data / code for other people to use
 - CITATION.cff on GitHub

<https://citation-file-format.github.io/>

```
cff-version: 1.1.0
message: If you use this software, please cite it as
below.
authors:
  - family-names: Druskat
    given-names: Stephan
    orcid: https://orcid.org/0000-0003-4925-7248
title: "My Research Software"
version: 2.0.4
doi: 10.5281/zenodo.1234
date-released: 2017-12-18
```

Proper credit, con't.

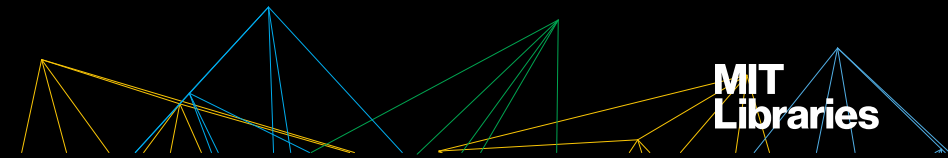
- List your license
- Things aren't open source just because they are available openly!
- License specifies how the work can be used going forward

Instructional Material

All Software Carpentry, Data Carpentry, and Library Carpentry instructional material is made available under the [Creative Commons Attribution license](#). The following is a human-readable summary of (and not a substitute for) the [full legal text of the CC BY 4.0 license](#).

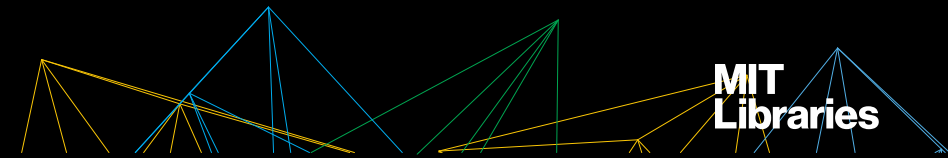
You are free:

- to **Share**---copy and redistribute the material in any medium or format
- to **Adapt**---remix, transform, and build upon the material



Final checklist!

- ❑ Title, who the data/code was created by and when
- ❑ Funding info, links to other resources
- ❑ What the data/code is for and why it's useful
- ❑ What the components are
- ❑ How to install / use it
- ❑ Citation and license



Variations on a theme

- The “data nutrition label” and the “model card” for ML datasets
- Documentation generators

<https://arxiv.org/abs/1803.09010>

Movie Review Polarity

Thumbs Up? Sentiment Classification using Machine Learning

Motivation

For what purpose was the dataset created? Was there a specific task in mind? Was there a specific gap that needed to be filled? Please provide a description.

The dataset was created to enable research on predicting sentiment polarity—i.e., given a piece of English text, predict whether it has a positive or negative affect—or stance—toward its topic. The dataset was created intentionally with that task in mind, focusing on movie reviews as a place where affect/sentiment is frequently expressed.¹

Who created the dataset (e.g., which team, research group) and on behalf of which entity (e.g., company, institution, organization)?

The dataset was created by Bo Pang and Lillian Lee at Cornell University.

Who funded the creation of the dataset? If there is an associated grant, please provide the name of the grantor and the grant name and number.

Funding was provided from five distinct sources: the National Science Foundation, the Department of the Interior, the National Business Center, Cornell University, and the Sloan Foundation.

Any other comments?

None.

Composition

What do the instances that comprise the dataset represent (e.g., documents, photos, people, countries)? Are there multiple types of instances (e.g., movies, users, and ratings; people and interactions between them; nodes and edges)? Please provide a description.

The instances are movie reviews extracted from newsgroup postings, together with a sentiment polarity rating for whether the text corresponds to a review with a rating that is either strongly positive (high number of stars) or strongly negative (low number of stars). The sentiment polarity rating is binary {positive, negative}. An example instance is shown in figure 1.

these are words that could be used to describe the emotion characters in his latest, limbo . but no , i use them to describe sitting through his latest little exercise in indie egomania . i things . but using some hackneyed , whacked-out , screwing on a movie is unforgivable . i walked a half-mile through two hours of typical , plodding sayles melodrama to complete and total copout finale . does sayles think he's roger

Figure 1. An example “negative polarity” instance, ta neg/cv452.tok-18656.txt.

exception that no more than 40 posts by a single cluded (see “Collection Process” below). No test determine representativeness.

What data does each instance consist of? “Raw” processed text or images/or features? In either case, please description.

Each instance consists of the text associated with obvious ratings information removed from that text were found and later fixed). The text was down-cased tags were removed. Boilerplate newsgroup header removed. Some additional unspecified automated done. Each instance also has an associated target value (+1) or negative (-1) sentiment polarity rating based on the number of stars that that review gave (details on the number of stars to polarity is given below in “Details”).

Is there a label or target associated with each instance provide a description.

The label is the positive/negative sentiment polarity from the star rating, as described above.

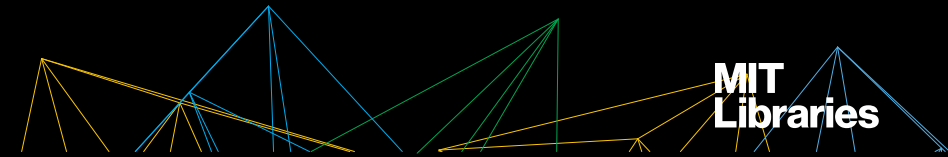
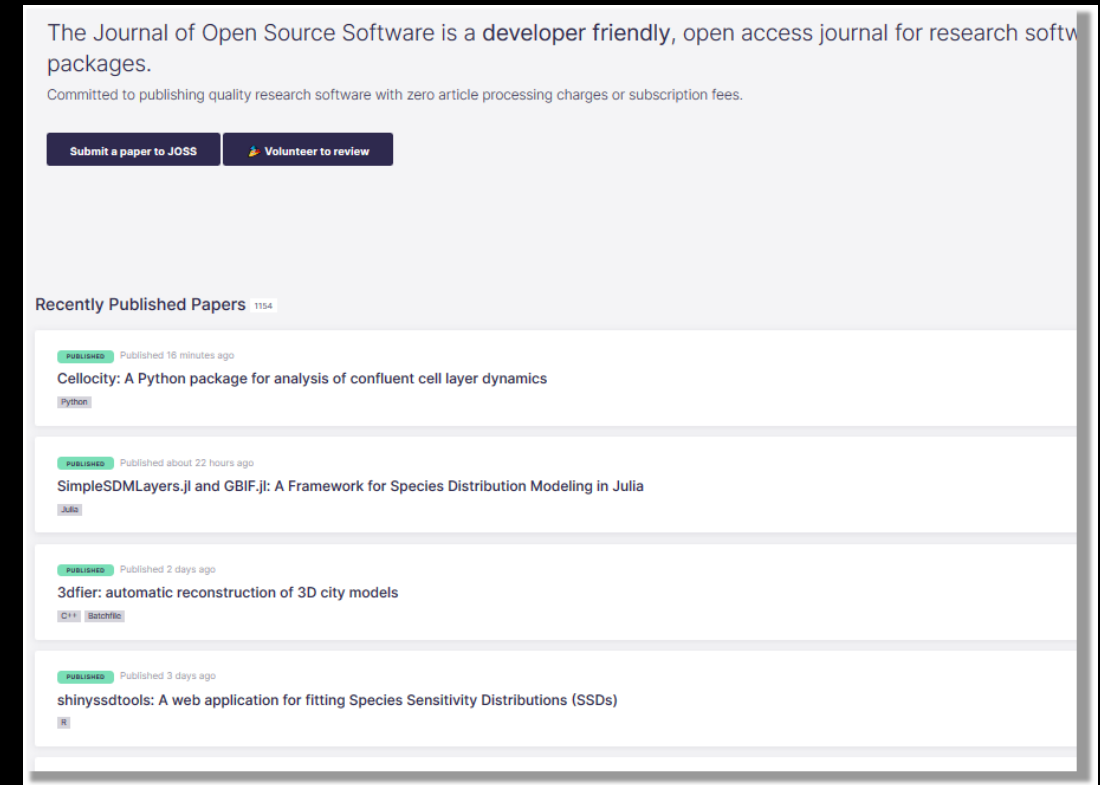
Is any information missing from individual instances provide a description, explaining why this information is cause it was unavailable). This does not include information, but might include, e.g., redacted text.

Everything is included. No data is missing.

Are relationships between individual instances maintained?

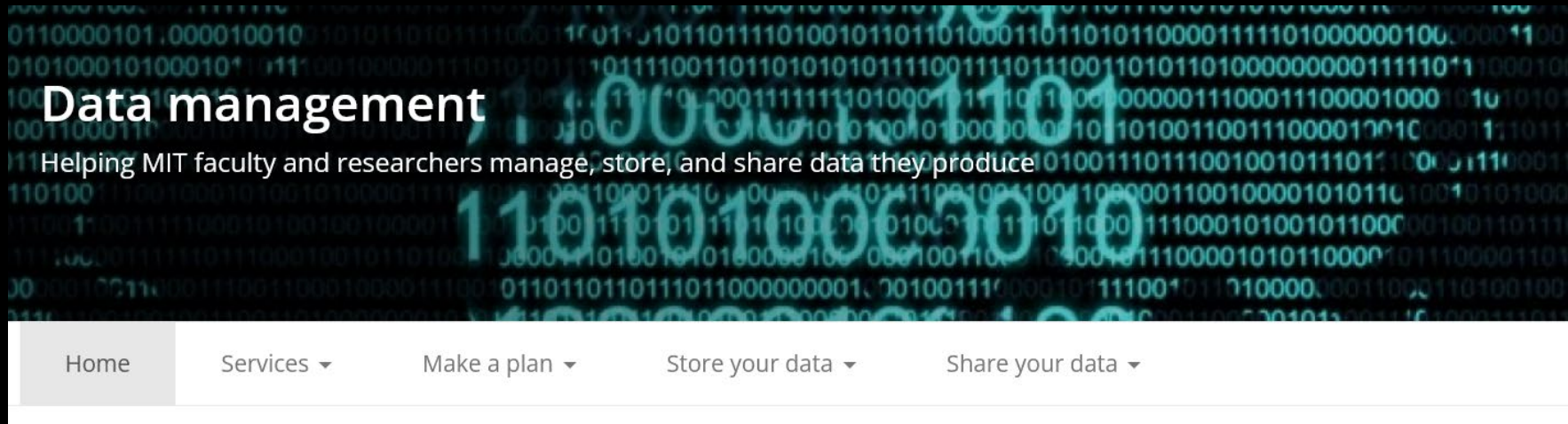
Variations, con't

- Software papers and data papers
- Gives you a citation that can be used *in addition to* citing the software itself - good for credit!
- Should include basic details and methods
- Example:
 - VanderPlas, “mst_clustering: Clustering via Euclidean Minimum Spanning Trees”
<https://www.theoj.org/joss-papers/joss.00012/10.21105.joss.00012.pdf>



Get help!

- Research Data Management services at MIT Libraries:
- <https://libraries.mit.edu/data-management>
- The rest of our classes: <https://libraries.mit.edu/news>
- data-management@mit.edu



Sources & Resources

Please respect the copyrights and licenses of the creators

Templates:

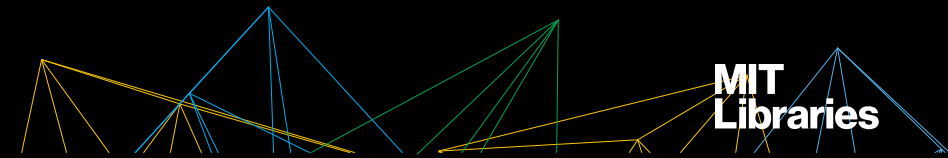
- <https://www.makeareadme.com/>
- https://social-science-data-editors.github.io/template_README/template_README.html

Data READMEs:

- https://datadryad.org/stash/best_practices#describe
- <https://data.research.cornell.edu/content/readme>

Code READMEs:

- <https://medium.com/@meakaakka/a-beginners-guide-to-writing-a-kickass-readme-7ac01da88ab3>
- <https://docs.github.com/en/repositories/managing-your-repositorys-settings-and-features/customizing-your-repository/about-readmes>
- <https://github.com/18F/open-source-guide/blob/18f-pages/pages/making-readmes-readable.md>



Sources & Resources

Citations and licenses:

- <https://research-software.org/citation/developers>
- CFF project: <https://citation-file-format.github.io/>
- Codemeta project: <https://codemeta.github.io/>
- <https://www.dcc.ac.uk/guidance/how-guides/cite-datasets>
- <https://choosealicense.com/>
-

Software papers and data papers:

- Software papers: <https://www.software.ac.uk/which-journals-should-i-publish-my-software>
- Many venues for data papers! Contact Data Management data-management@mit.edu for help identifying some

Other formats:

- See “Datasheets for Datasets” (Gebru et al) <https://arxiv.org/abs/1803.09010>
- “Model Cards for Model Reporting” (Mitchell et al) <https://arxiv.org/abs/1810.03993>
- The Data Nutrition Project <https://datanutrition.org/>